

End Semester Exam: Real Analysis

College For Integrated Studies
University of Hyderabad

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Duration: 3 Hours
Maximum Score: 60 points

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Course Code: MM 263

Instructions: Answers without justification will receive a score of zero.

1. (a) If $S := \left\{ \frac{1}{n} - \frac{1}{m} : m, n \in \mathbb{N} \right\}$. Find $\inf S$ and $\sup S$. [4]

(b) Let S be a nonempty bounded set in $\mathbb{R}$ and $a < 0$. Show that $\sup \{aS\} = a \inf \{S\}$, [4]

2. Let $\{x_n : n \in \mathbb{N}\}$ be an arbitrary collection of **non-negative** real numbers such that $\sum_{n=1}^{\infty} x_n$ converges. Examine whether following series converge / diverge / inconclusive.

(a) $\sum_{n=1}^{\infty} x_n^2$ [4]

(b) $\sum_{n=1}^{\infty} \sqrt{x_n}$ [4]

3. (a) Let $0 < r < 1$. Assume that (x_n) satisfies the following condition:

$$|x_{n+2} - x_{n+1}| \leq r |x_{n+1} - x_n|, \quad \forall n \in \mathbb{N}.$$

Show that (x_n) is Cauchy sequence. [4]

(b) Show that following series converges and determine its sum:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

[4]

4. (a) Let D be a nonempty subset of $\mathbb{R}$ and let $f : D \rightarrow \mathbb{R}$. Write in terms of ε and δ the meaning of saying that f is not continuous at $x_0 \in D$.
(In other words, using ε and δ , complete the statement: f is not continuous at $x_0 \in D$ iff)

[4]

- (b) Show that every polynomial of odd degree with real coefficients has at least one real root.

[4]

5. (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$

TRUE or FALSE: f is differentiable at 0.

[4]

- (b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable such that $f(0) = f(1) = 0$. Show that there exists $c \in (0, 1)$ such that $f'(c) = f(c)$.

[4]

6. (a) Prove / Disprove: $x - \frac{x^3}{3!} < \sin x < x - \frac{x^3}{3!} + \frac{x^5}{5!}$ for all $x \in (0, \pi)$

[4]

- (b) Prove / Disprove: $\lim_{x \rightarrow 0} \frac{e^{-\frac{1}{x^2}}}{x}$ does not exist.

[4]

7. (a) Find $f : [0, 1] \rightarrow \mathbb{R}$ such that $|f|$ is integrable but f is not integrable.

[4]

- (b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a bounded function. Then show that f is integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that

$$U(P, f) - L(P, f) < \varepsilon.$$

[4]

- (c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let $g(x) = \int_0^x (x-t)f(t) dt$ for all $x \in \mathbb{R}$. Show that $g''(x) = f(x)$ for all $x \in \mathbb{R}$.

[4]

All the best !